

13 Ans: A

$$pV = nRT$$

$$\frac{1}{p} = \frac{1}{nRT} V, \quad \text{hence, gradient} = \frac{1}{nRT}$$

$$\text{When } n \text{ and } T \text{ are both halved, gradient} = \frac{1}{\frac{n}{2} R \frac{T}{2}} = 4 \frac{1}{nRT},$$

Therefore the new gradient is 4 times that of the original graph.

14 Ans: D